



IMPerSumm: Information-Modulated User Preference Modeling for Personalized Text Summarization

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Abstract

Document summarization helps readers focus on the content-of-interest – a *highly subjective* and time-variant quantity, making *personalized summarization* essential. Prior state-of-the-art (SOTA) approaches either flatten evolving preferences into static personas, rely on LLM prompting that fails under long-horizon histories, or encode user interactions without explicit action types (*click*, *skip*, *summarize*, *click summary*), leaving preference dynamics under-modeled. To this end, we propose IMPerSumm, an information-modulated user preference encoder for personalized text summarization. IMPerSumm represents user histories as trajectories in a *user-interaction* graph. It dynamically modulates the mutual information between past interactions and new action-induced signals over a trajectory, and models short-term reactivity, long-term drift, and rare-event reinforcement using memory kernels. We empirically evaluate IMPerSumm on three benchmark datasets – PENS, OpenAI-Reddit, and PersonalSum using PerSEval, a personalization metric with strong human correlation. IMPerSumm outperforms SOTA personalized summarizers (PENS, GTP, Signature-Phrase) by 0.43 ↑ (on avg.), and strong LLM baselines (DeepSeek-R1-14B, LLaMA-2-13B, Mistral-7B, Zephyr-7B) by 0.37 ↑. Strikingly, we also observe that although the IMPerSumm encoder is trained on the personalized summarization task on the PENS dataset, it outperforms SOTA news recommendation systems on the standard *de-facto* MIND benchmark dataset (accuracy gain w.r.t best: 2.5/3.5/6.6 ↑ w.r.t MRR/nDCG@5/10). This shows that the IMPerSumm encoder develops cross-task capability of news recommendation.

CCS Concepts

• **Information systems** → **Summarization; Personalization; Recommender systems.**

Keywords

Personalized text summarization, User preference modeling, Temporal preference drift, User interaction graphs, Recommender systems



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Resource Availability:

The source code of this paper has been made publicly available at <https://doi.org/10.5281/zenodo.18324258>.

1 Introduction

With information overload, personalized summarization is vital for tailoring updates to a reader's specific interests, especially for multi-aspect documents catering to diverse topics [6]. Most existing methods rely on *static* persona traits [8, 15, 21]. However, preference datasets representing user reading behavior, like MS/CAS PENS [2], show that user interests shift over time at fine-grained sub-topic levels. This poses a challenge even for state-of-the-art (SOTA) LLMs, which underperform when detailed user-interaction histories are injected in prompts for in-context personalized summarization [25, 32].

Recent models aim to *explicitly* represent the evolving user interaction (i.e., *click*) trajectories. However, they fail to model either the temporal ordering of the user-interaction sequence or the nuances of the other user-interactions (e.g. *skip*, *summarize*, etc.) that affect the information content flow over the trajectory [2, 22, 29, 36, 43]. This is empirically shown from the benchmarking by Dasgupta et al. [6] w.r.t. PerSEval, the only personalized summarization metric with strong human-judgment correlation.

To address this gap, we propose IMPerSumm, an encoder-decoder-based personalized summarizer that represents user reading history as an *information-content* (IC) flow modulated by user actions (*click* vs. *skip*). At each timestep, the IMPerSumm encoder aligns the current document's IC with the accumulated history, estimates the document novelty (thereby, new interest) using a Kernel Density Estimation (KDE)-based distributional surprise model, and modulates the flow accordingly. A multi-layer perceptron (MLP)-based header predicts the next behavior embedding that entangles the query document and the corresponding latent summary. The decoder then disentangles the latent summary embedding and generates a contextualized query embedding. This is then fed into a pre-trained frozen *vanilla* summary-decoder for the final personalized summary.

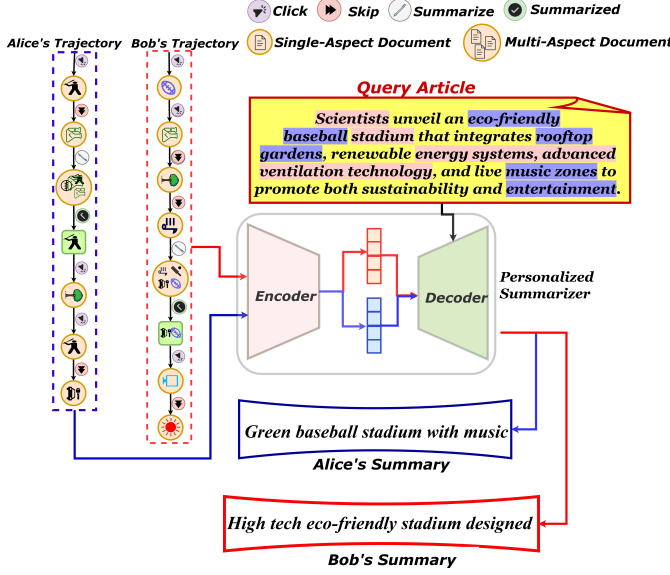


Figure 1: Personalized Summarization Pipeline: A behavior-aware encoder–decoder encodes each user’s trajectory as the preference condition to generate personalized summaries of the same query article.

We pose three core research questions to investigate IMPerSumm’s effectiveness as an end-to-end personalized summarizer:

- **RQ-1:** To what extent does explicit action-specific user encoding affect personalized summarization quality (w.r.t PSE) compared to SOTA RNN and Transformer-based encoders?
- **RQ-2:** Beyond encoder-based modeling, can stronger personalization emerge either from history-prompted mid-size LLMs or from oracle-style summarizers with access to gold references?
- **RQ-3:** How well does IMPerSumm’s encoder trained for personalized summarization on news datasets (PENS and PersonalSum) generalize to the related task of personalized recommendation, compared to SOTA news recommendation models?

To address RQ-1, we use PENS [2], Norwegian PersonalSum [50], and OpenAI Reddit [41] datasets. We benchmark the PENS summarization framework (augmented with NAML [43], EBNR [29], NRMS [44] user encoder), GTP [36] (using the TrRMio encoder) and Signature-Phrase [3], and observe that IMPerSumm yields **0.49/0.48/0.5** w.r.t PSE-JSD/SU4/METEOR. For RQ-2, we evaluate Zephyr-7B [38], LLaMa2-13B [37], Mistral-7B [19], and DeepSeek-R1-14B [7] as strong baselines. We observe that IMPerSumm outperforms the best baseline (DeepSeek-R1) by **0.27/0.41/0.42**. We also follow Patel et al. [32] to evaluate IMPerSumm against oracle BigBird-Pegasus [49] and SimCLS [27] (having access to gold references) – IMPerSumm outperforms by **0.32**. For RQ-3, we evaluate the cross-task capability of the PENS-trained IMPerSumm-encoder by comparing against 30 strong news recommendation baselines on the standard, widely adopted MIND-large news recommendation dataset. We find that IMPerSumm’s encoder surpasses the best baseline FastFormer-PLM-NR [45] by **3.5/6.6** in terms of nDCG@5&@10 and **2.5** w.r.t. MRR.

2 Related Work

Personalized Summarization Evaluation: Personalized summarization aims to align generated summaries with a user’s subjective expectations, as signalled by the evolving reading behavior (clicks, skips, summarize interactions). Traditional accuracy metrics fail to capture this personalization, as shown by Vansh et al. [40], who proposed the EGISSES metric. PerSeval [6] refines EGISSES by penalizing accuracy drops. We adopt PerSeval to evaluate our proposed IMPerSumm summarizer.

Training/Evaluation Datasets: Effective personalization demands datasets with (i) temporally-ordered user interactions, (ii) user-specific gold summaries for shared documents, and (iii) diverse, shifting topics. Unlike CNN/DM or MultiNews [9, 17], PENS [2] and translated PersonalSum [50] satisfy these. We use both for our experiments, and also OpenAI-Reddit [41] to synthetically derive temporal orders for personalized summarization. PENS logs clicks/skips and summaries per user, averaging 13.6 topics and 52.83 sub-topics, with topic change rates of 0.77/0.93 per step. Its fine-grained behavioral diversity makes it a standard benchmark [22, 36, 48].

Personalized Summarizers: Most personalized summarizers assume a *static* user persona guided summarization (e.g., GSUM, CTRLSum, TMWIN, Tri-Agent) [8, 15, 20, 47]. On the other hand, models such as PENS augment external *dynamic* user-trajectory encoders such as NRMS, NAML, and EBNR [2, 29, 43, 44]. Others, such as GTP [36], generate control but omit temporal distinctions. SCAPE integrates content and stylistic features [22]. However, none capture session-aware, action-specific dynamics. Although prompting LLMs with preference history, termed *In-Context-Personalization Learning (ICPL)* [32], outperforms these models, it fails in practice. This is because most LLMs cannot handle long, complex prompts and suffer the “*lost-in-the-middle*” effect – i.e., they ignore important trajectory segments that are in the middle [11, 25, 33]. Moreover, richer preference cues can degrade personalization, illustrating the iCOPERNICUS “*Paradox of Less is More*” [32]. To capture action-specific dynamics, IMPerSumm leverages MI-based relevance [5, 39, 51].

Personalized Summarization and Recommendation. While personalized summarization and recommendation have long evolved in parallel, recent work highlights their convergence. The PENS framework shows that user embeddings from recommendation models can guide abstractive summarization, while SumRecom [12] casts summarization itself as a recommendation task via user feedback. Conversely, summarization enhances recommendation by generating explanatory profiles, as in PGHIS [26]. Multi-task approaches such as P5 [52] further demonstrate that user preference signals transfer across retrieval, ranking, and generation. Collectively, these works point to a shared foundation in modeling user histories, with different outputs (ranked items vs. generated summaries). By enabling summarization-to-recommendation transfer without retraining, IMPerSumm advances this line toward unified user models bridging generative and predictive personalization.

3 Personalized Summarization: Formulation

In this direction, we introduce the User-Interaction-Graph (UIG), a data model for dynamic behavior trajectories.

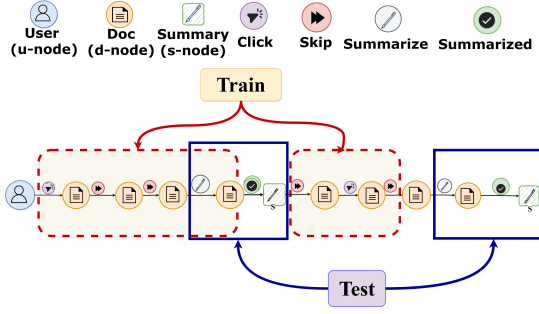


Figure 2: User-Interaction Graph construction linking u-, d- and s-nodes through action edges (click, skip, summarize) over training and test partitions; see Appendix C for details.

3.1 Preference Data Modeling

We represent user histories as a **User-Interaction Graph (UIG)**, a directed acyclic graph $G = \langle N, E \rangle$, where the node set N consists of three disjoint types: (i) **u-nodes** $u^{(t_0)}$ denoting a user at initial timestep t_0 , (ii) **d-nodes** $d^{(t_p)}$ representing documents interacted at timestep t_p , and (iii) **s-nodes** $s_j^{(t_q)}$ representing user-specific summaries requested or generated at time t_q for a document viewed at t_{q-1} . The edge set E encodes user actions: $a_d^{(t_p)} \in \{\text{click}, \text{skip}, \text{summarize}\}$ on documents, and $a_s^{(t_q)}$ as the follow-up *sumGen* action connecting a document $d^{(t_{q-1})}$. A user **trajectory** τ_u is then a time-ordered sequence of such interactions, beginning at $u^{(t_0)}$. Each trajectory can be decomposed into **behavior triples** $b_{u_j}^{(t_i)} : \langle hd^{(t_{i-1})}, a^{(t_i)}, tl^{(t_i)} \rangle$, denoting the action-triggered transition connecting the *hd*-node $hd^{(t_{i-1})}$ and *tl*-node $tl^{(t_i)}$. Such triples may represent a *document-to-document*, *document-to-summary*, and *summary-to-document* transitions. The UIG $\mathcal{T}$ is a pool of trajectories, used as $\mathcal{T}_{\text{train}}$ for training and $\mathcal{T}_{\text{test}}$ for evaluation. Hence, a UIG is a dynamic temporal knowledge graph (TKG) of user behavior (UIG construction: Figure 2; Appendix. C).

While UIG captures rich temporal detail, directly modeling its raw structure quickly becomes computationally expensive and noisy over long horizons. This is particularly challenging in personalized summarization, where fine-grained shifts in preference must be retained without overwhelming the model. Recent work in sequential recommendation suggests that *hierarchical abstractions* improve long-horizon accuracy by condensing low-level interactions into higher-order behavioral units [4, 30, 31, 53, 55]. Motivated by this, we introduce a bi-level UIG: each behavior triple $b_{u_j}^{(t_i)}$ is a **b-node** in the higher-level **b-tier** $\tau_b^{u_j}$ constructed on top of the **u-tier** $\tau_u^{u_j}$, linked by **nextBehavior** edges. Task-1 (next b-node prediction) operates on the b-tier. We construct u-tier and b-tier trajectories from PENS [2], English PersonalSum [50], and OpenAI (Reddit) [41] as $\mathcal{T}_{\text{train}}^{\text{P}}$, $\mathcal{T}_{\text{test}}^{\text{P}}$, $\mathcal{T}_{\text{train}}^{\text{PS-EN}}$, $\mathcal{T}_{\text{test}}^{\text{PS-EN}}$, and $\mathcal{T}_{\text{train}}^{\text{OAI}}$, $\mathcal{T}_{\text{test}}^{\text{OAI}}$.

Problem Formulation. We formulate the personalized summarization task into three stages: **Task 1** - predict the next behavior triple $b_{(q,u_j)}$ (i.e., embedding) from τ^{u_j} embedding and d_q embedding as user's next interest, **Task 2** - decode $b_{(q,u_j)}$ to extract the latent

representation of $s_{(q,u_j)}$, and **Task 3** - generate personalized summary using a frozen pre-trained summarizer decoder by feeding in $s_{(q,u_j)}$ embedding contextualized by d_q embedding.

4 Approach

4.1 Preference as Information Flow Modulation

A central challenge in modeling evolving user preferences is balancing **continuity with the past** against **adaptation to novelty**. When a user clicks, skips, or requests a summary, the model must decide: *does this reinforce existing interests, or mark a shift in focus?* We frame this as a problem of **information flow modulation**, where each new interaction is judged by how much information it shares with accumulated history. Mutual Information (MI) provides a principled measure of such dependency: high MI implies continuity with prior interests, while low MI suggests novelty or drift. Since preference signals reside in high-dimensional embedding spaces, MI is estimated using **kernel density estimation (KDE)** [35, 54], which smooths distributions over interaction embeddings to compute alignment between current and historical representations. Preference evolution, however, unfolds across **multiple timescales**. Short-term kernels capture immediate reactivity, long-term kernels encode gradual drift, and rare-event kernels highlight infrequent but decisive shifts. Moreover, not all actions contribute equally: skips convey weaker or negative signals, clicks reinforce engagement, while summary requests mark deeper intent. In this way, a user trajectory τ^{u_j} is modeled as a sequence of **regressive information updates**, where timescale-specific kernels offer complementary views of history and action-specific gates modulate their strength. This yields a dynamic, information-theoretic account of preference evolution sensitive to both *when* an action occurs and *how* it is expressed. Section 4.2 instantiates these principles within the IMPerSumm architecture.

4.2 The IMPerSumm Model

Building on preference-as-information-flow modulation, we propose IMPerSumm as an encoder-decoder instantiation. IMPerSumm addresses three tasks defined in Section 3.1 sequentially using an encoder for task 1, a decoder contextualizer for task 2, and a frozen pre-trained T5-decoder for task 3. The end-to-end pipeline is shown in Figure 3 (Notations: Table 7 in Appendix).

4.2.1 Task 1: Next b-node Prediction. We now detail how IMPerSumm encoder models $\tau_b^{u_j}$ via a regressive information modulation cell (Section 4.1), producing a high-dimensional embedding for $\tau_b^{u_j}$.

Initialization of u-tier. To enable Task-1 on the b-tier, we initialize the user's u-tier trajectory τ^{u_j} by embedding each document and summary node using the E5 model [42]. For each behavior triple $b_{u_j}^{(t_i)}$, the head and tail embeddings are $e_{hd}^{(t_{i-1})}$ and $e_{tl}^{(t_i)}$, respectively. The initial u-node embedding $e_{u_j}^{(t_0)}$ derives from the title embedding of $d^{(t_0)}$, addressing cold starts. This seeds the user's starting information state, anchored in the first document's topical semantics before any behavior-driven modulation.

b-tier Architecture. IMPerSumm's b-tier encoder models the evolving $\tau_b^{u_j}$ using a recurrent architecture of **b-cells**, each of which

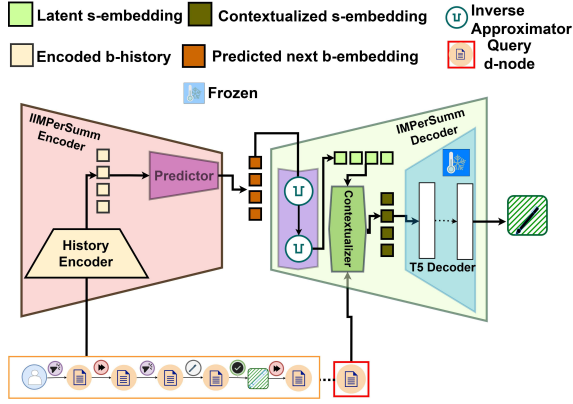


Figure 3: IMPerSumm End-to-end pipeline diagram: User trajectory is encoded by History Encoder; Predictor estimates the next behavior node; Query document, contextualized by extracted latent summary, is fed into (frozen) T5-decoder for personalized summary.

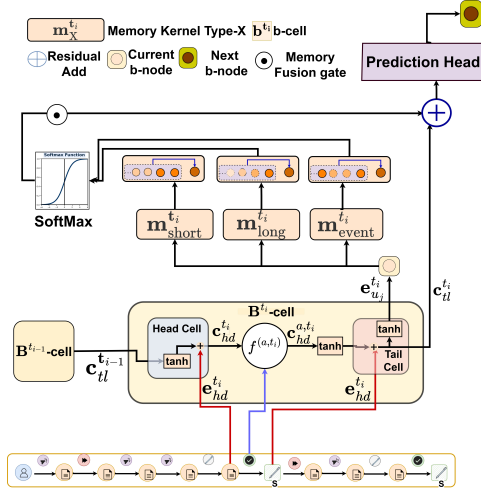


Figure 4: IMPerSumm Encoder: A chain of b-cells (Head & Tail cells), each representing a b-node of the underlying UIG (see Section 3.1), form b-tier; Information Content Flow is modulated in the b-cell based on user action to generate b-node embedding; Memory Functions (w/ kernels) further modulate b-node embedding; Prediction head predicts next b-node.

produces a *b*-node embedding ($\mathbf{e}_{u_j}^{(t_i)}$) at timestep t_i . Each *b*-cell integrates user interactions through three sequential components, (i) the **head-cell**, (ii) the **action-control**, and (iii) the **tail-cell**. The t_i -th head-cell fuses the incoming **behavior history** (i.e., the previous *b*-cell's **tail-cell content** $\mathbf{c}_{tl}^{(t_{i-1})}$) and the *hd*-node embedding $\mathbf{e}_{hd}^{(t_{i-1})}$ to generate the **head-cell content** $\mathbf{c}_{hd}^{(t_i)}$ as follows (W_h, W_{hd} are learnable): $\mathbf{c}_{hd}^{(t_i)} = \tanh(W_h \cdot \mathbf{c}_{tl}^{(t_{i-1})} + W_{hd} \cdot \mathbf{e}_{hd}^{(t_{i-1})})$. The action-control (a), representing one of the four possible transition actions, then adjusts the $\mathbf{c}_{hd}^{(t_i)}$, and modulates what information is to be retained, enhanced, or discarded using an information-alignment

function $f_{\sigma}^{(a, t_i)}$ as: $\mathbf{c}_{hd}^{(a, t_i)} = f_{\sigma}^{(a, t_i)}(\mathbf{c}_{hd}^{(t_i)}, \mathbf{e}_{tl}^{(t_i)})$. The tail-cell then combines the action-modulated $\mathbf{c}_{hd}^{(a, t_i)}$ with the tail-node embedding $\mathbf{e}_{tl}^{(t_i)}$ to produce the **tail-cell content** $\mathbf{c}_{tl}^{(t_i)}$ and corresponding **b-node embedding** $\mathbf{e}_{u_j}^{(t_i)}$ as: $\mathbf{c}_{tl}^{(t_i)} = \tanh(W_a \cdot (\mathbf{c}_{hd}^{(a, t_i)} + \mathbf{e}_{tl}^{(t_i)}))$; $\mathbf{e}_{u_j}^{(t_i)} = \tanh(W_b \cdot \mathbf{c}_{tl}^{(t_i)} + \mathbf{b}_b)$.

Base Encoder. In its simplest form, the **information-alignment function** $f_{\sigma}^{(a, t_i)}$ is a cosine similarity, such that $\mathbf{c}_{hd}^{(a, t_i)} = \lambda_c \cdot f_{\cos}(\mathbf{c}_{hd}^{(t_i)}, \mathbf{e}_{tl}^{(t_i)})$, where λ_c is a learnable influence factor to capture the influence of every alignment signal. However, embedding similarity (such as cosine) lacks the ability to specifically capture how much information is carried over *in contrast to new information*. Moreover, it does not capture the nuances of action-specific updates explicitly for *clicks*, *skips*, or *summarize*.

MI (Mutual Information) Encoder. To capture nuanced, action-specific information flow (or modulation) across each *b*-cell, we model the information-alignment function $f_{\sigma}^{(a, t_i)}$ as **KDE-based pair-wise mutual information modulation** (see Section 4.1) as: $f_{\sigma}^{(a, t_i)} = I(tl^{(t_i)}; hd^{(t_{i-1})}) = I_K(\mathbf{e}_{tl}^{(t_i)}; \mathbf{c}_{hd}^{(t_{i-1})})$, where $I(\cdot; \cdot)$ denotes mutual information. High I indicates continuity with past preferences, while low I indicates novelty or drift. In practice, we estimate I using a **kernelized estimator**: $I_K(\mathbf{e}_{tl}^{(t_i)}; \mathbf{c}_{hd}^{(t_{i-1})})$. The marginals and joints are obtained via **Gaussian kernel density estimation (KDE)** [54]. Given cumulative history embeddings $\{\mathbf{c}_{hd}^{(t_j)}\}_{j=1}^{i-1}$, the marginal density is

$$\hat{p}(\mathbf{e}_{tl}^{(t_i)}) = \frac{1}{(i-1)\lambda} \sum_{j=1}^{i-1} K\left(\frac{\mathbf{e}_{tl}^{(t_i)} - \mathbf{c}_{hd}^{(t_j)}}{\lambda}\right),$$

where $K(\cdot)$ is the Gaussian kernel and λ the bandwidth. Here $(i-1)$ is the number of past interactions. Smaller λ emphasizes fine-grained alignment with recent history, larger λ smooths over longer dependencies¹. The modulated head-cell content $\mathbf{c}_{hd}^{(clk, t_i)}$ of the *click* action is generated using the learnable *click-gate* $g^{(clk, t_i)}$ that depends on the (normalized) time spent (called *dwell time*; δ_{dwell}) by the user on any *d*-node. In other words, if a user spends less time on a node, then the information content of the *b*-cell should decrease. $\mathbf{c}_{hd}^{(clk, t_i)}$ is computed as: $\mathbf{c}_{hd}^{(clk, t_i)} = g^{(clk, t_i)} \odot I_K(\mathbf{e}_{tl}^{(t_i)}; \mathbf{c}_{hd}^{(t_{i-1})})$, where: $g^{(clk, t_i)} = \frac{\delta_{dwell}^{t_i}}{\max_{j=1:(N-U)} \delta_j} \cdot \tanh(W_g^{clk} \cdot \mathbf{e}_{tl}^{(t_i)}) \odot \mathbf{c}_{hd}^{(skp, t_i)}$ for the *skip* action is generated using the learnable *skip-gate* $g^{(skp, t_i)}$. Since *skip* denotes user's disinterest in the content of the tail-node to certain degree, $g^{(skp, t_i)}$ determines *how much of $\mathbf{e}_{tl}^{(t_i)}$ is to be suppressed* (i.e., overlapping information with the incoming history should not be erased), and *how much of the inflowing history is to be retained* (i.e., redundant information in the history needs to be erased). $\mathbf{c}_{hd}^{(skp, t_i)}$ is computed as:

$$\mathbf{c}_{hd}^{(skp, t_i)} = g^{(skp, t_i)} \odot \mathbf{e}_{tl}^{(t_i)} + (1 - g^{(skp, t_i)}) \odot I_K(\mathbf{e}_{tl}^{(t_i)}; \mathbf{c}_{hd}^{(t_{i-1})}), \quad (1)$$

where $g^{(skp, t_i)} = \tanh(W_g^{skp} \cdot \mathbf{e}_{tl}^{(t_i)})$. The *summarize* command action signals a *focused intent* of the user. Thus, the information is modulated via a *focus-gate* $g^{(fc, t_i)}$. $g^{(fc, t_i)}$ determining the information likely to be salient (contained in $tl^{(t_i)}$) if clicked. Since the

¹We set $\lambda = 0.6$ via Silverman's rule-of-thumb [35], $\lambda = 1.06 \hat{\sigma} n^{-1/5}$, where $\hat{\sigma}$ is the sample standard deviation and n the sample size.

summarize command comes with an interest developed by the **title** of the document, $g^{(fc, ti)}$ captures the *interest in the title* of the d -node to be summarized, with the *subjective focus* (fc).

$$\mathbf{c}_{hd}^{(fc, ti)} = g^{(fc, ti)} \odot I_K(\mathbf{e}_{tl}^{(ti)}; \mathbf{c}_{hd}^{(ti)}), \quad (2)$$

where $g^{(fc, ti)} = \tanh(W_g^{fc} \cdot \mathbf{e}_{TITLE(tl)}^{(ti)})$. Each *summarize* is followed by an automated *summGen* transition-action, which generates the user's expected summary s^{uj} for the document d^{ti} from the u -tier. Precisely, $\mathbf{c}_{hd}^{(summ, ti)}$ is generated as a linear combination of two learnable gates: *interest-gate* $g^{(summ, ti)}$, and *accuracy-gate* $g^{(acc, ti)}$. The *interest-gate* $g^{(summ, ti)}$ captures the history-influenced interest of the user, determining the content alignment of the document, and *accuracy-gate* $g^{(acc, ti)}$ ensures the alignment of expected summary s^{uj} with document d^{ti} , ensuring the summary-content to be faithful to the document as:

$$\mathbf{c}_{hd}^{(summ, ti)} = g^{(summ, ti)} \odot I_K(\mathbf{e}_{tl}^{(ti)}; \mathbf{c}_{hd}^{(ti)}) + g^{(acc, ti)} \odot I_K(\mathbf{e}_{tl}^{(ti)}; \mathbf{e}_{hd}^{(ti)}) \quad (3)$$

where $g^{(summ, ti)} = \tanh(W_g^{summ} \cdot \mathbf{e}_{tl}^{(ti)})$ and $g^{(acc, ti)} = \tanh(W_g^{acc} \cdot \mathbf{e}_{hd}^{(ti)})$. Although $\mathbf{e}_{buj}^{(ti)}$ captures fine, action-specific updates at each timestep t_i , it's a local representation, biased to immediate interactions and weak at modeling long-term shifts or separating fleeting spikes from sustained themes.

Memory Kernel Augmentation. To align users' short-term current interest with that of the longer-term persistent interests, we propose a multi-scale memory mechanism inspired by hierarchical caches [16] and kernelized state-space models like S4 and Mamba [13, 14]. These works affirm that fusing short-term reactivity with long-term continuity improves sequential modeling. Unlike hierarchical memory slots or implicit ODE states, we use three types of temporally kernelized (via $\mathcal{K}$ kernels) memory functions ($\mathbf{m}^{(ti)}$): (i) short-term, (ii) long-term, and (iii) event-specific. $\mathbf{m}^{(ti)}$ enables the past b -node embeddings $\{\mathbf{e}_{buj}^{(tk)}\}_{k=1:i-1}$ to shape the present b -node embeddings $\mathbf{e}_{buj}^{(ti)}$. The short-term memory ($\mathbf{m}_{short}^{(ti)}$) captures recent, *reactive* behavior patterns as: $\mathbf{m}_{short}^{(ti)} = \sum_{j=1}^{i-1} \mathcal{K}_{short}^{tj} \odot \mathbf{e}_{buj}^{(tj)}; \mathcal{K}_{short}^{tj} = \exp(-W_{short} \cdot \mathbf{e}_{buj}^{(tj)})$, where $\mathcal{K}_{short}$ encodes Markovian recency by

assigning exponentially decaying weights. $\mathbf{m}_{short}^{(ti)}$ is ideal for fast-changing interests (e.g., interest spike in soccer articles during the World Cup). However, it forgets stable preferences (e.g., a basketball lover who returns to basketball articles week after week). To address this, the long-term memory function $\mathbf{m}_{long}^{(ti)}$ models *persistent* user interests as: $\mathbf{m}_{long}^{(ti)} = \sum_{j=1}^{i-1} \mathcal{K}_{long}^{tj} \odot \mathbf{e}_{buj}^{(tj)}; \mathcal{K}_{long}^{tj} = \exp\left(\frac{-W_{long}}{\|W_{short}\|} \cdot \mathbf{e}_{buj}^{(tj)}\right)$ with $\mathcal{K}_{long}$ is modeled by W_{long} , and scaled down by norm of W_{short} to ensure slower decay than $\mathcal{K}_{short}$. Thus, $\mathbf{m}_{long}^{(ti)}$ captures drift-aware, reinforced preferences (e.g., recurring basketball reads). However, it can miss rare yet impactful deviations (e.g., a sudden interest in chess due to some viral news). Hence, event-specific memory function is added as: $\mathbf{m}_{event}^{(ti)} = \sum_{j=1}^{i-1} \mathcal{K}_{event}^{tj} \odot \mathbf{e}_{buj}^{(tj)}; \mathcal{K}_{event}^{tj} = \exp(-(W_{event} \cdot \mathbf{e}_{buj}^{(tj)})^\top \mathbf{e}_{buj}^{(ti)})$. $\mathbf{m}_{event}^{(ti)}$ highlights isolated yet meaningful events-interactions that are far in the past but highly influential to current user intent. W_{event} transforms a past b -node embedding.

The dot product with current b -node embedding $\mathbf{e}_{buj}^{(ti)}$ computes the affinity score. This score quantifies the impact of a past isolated behavior on the present.

Adaptive Memory Fusion (AMF). We now fuse all three memory functions to form the **memory-injected b -node embedding** ($\mathbf{m}_{uj}^{(ti)}$). However, the fusion needs to be adaptive because not all the memory functions (and features) should be equally activated at every step of τ^{uj} . To address this, we first construct a **Memory Matrix** $\mathbf{M}^{(ti)} = \text{SoftMax}([\mathbf{m}_{short}^{(ti)}; \mathbf{m}_{long}^{(ti)}; \mathbf{m}_{event}^{(ti)}]^\top)$ that contains the normalized weighted (in %) memory feature across all three types. We then pool the most important (i.e., maximum weight) features across the three memory functions to create the fused **memory vector** $\mathbf{m}_{fuse}^{(ti)} = \text{MaxPool}_{j \in \{\text{short}, \text{long}, \text{event}\}} \mathbf{M}_{i,j}^{(ti)}$. While $\mathbf{m}_{fuse}$ extracts only the important features (count: 3) across the memory functions at any time-step, it still gives equal importance to all the functions. To address this, we further apply a *fusion-gate* g_{fuse} to generate the corresponding **gated memory vector** $\mathbf{m}_{uj}^{(ti)} = g_{fuse}^{(ti)} \odot \mathbf{m}_{fuse}^{(ti)}$, where $g_{fuse}^{(ti)} = \tanh(W_{fuse} \cdot \mathbf{e}_{buj}^{(tj)})$. We add a residual $\mathbf{c}_{tl}^{(ti)}$ to $\mathbf{m}_{uj}^{(ti)}$ to

generate the final memory-infused b -node embedding $\mathbf{z}_b^{(ti)}$.

Next b -node Prediction. Once the user history (trajectory of l time-steps) is encoded in $\mathbf{c}_{tl}^{(ti)}$ (with the final b -node embedding $\mathbf{z}_b^{(tl)}$), we apply a next b -node embedding prediction header W_{next} on it to predict the query b -node at t_{l+1} as $\mathbf{z}_b^{(t_{l+1})} = W_{next} \cdot \text{ReLU}(W_z \cdot \mathbf{z}_b^{(tl)} + \mathbf{b}_{next})$.

Encoder Training. The encoder training objective ($\mathcal{L}_{enc}$) is a linear combination of two loss functions: *Alignment Loss* ($\mathcal{L}_{align}$) and the *Prediction Loss* ($\mathcal{L}_{next}$) as $\mathcal{L}_{enc} = \alpha \cdot \mathcal{L}_{align} + (1 - \alpha) \cdot \mathcal{L}_{next}$. $\mathcal{L}_{align}$ makes sure that during the next-node (i.e., $\mathbf{z}_b^{(t_{l+1})}$) prediction, the encoder maintains its original positional alignment at every step on τ_l^{uj} . Specifically, at each timestep t_i , a learnable position classifier W_{pos} on $\mathbf{z}_b^{(ti)}$ to generate positional probability distribution $\hat{\mathbf{p}}_b^{(ti)}$ over behavior node index-vocabulary B as $\hat{\mathbf{p}}_b^{(ti)} = \text{SoftMax}(W_{pos} \cdot \mathbf{z}_b^{(ti)})$. The total alignment loss is $\mathcal{L}_{align} = -\frac{1}{l} \sum_{i=1}^l \log \hat{\mathbf{p}}_b^{(ti)}$. $\mathcal{L}_{next}$ over behavior node vocabulary B is $-\log \hat{\mathbf{p}}_b^{(t_{l+1})}$, ensuring that the encoder learns to predict the next behavior of the user.

4.2.2 Task 2: Latent Summary Contextualization. The query b -node embedding $\mathbf{z}_b^{(t_{l+1})}$ predicted by the encoder represents a memory-infused entanglement of the behavior triple $\langle d_q, \text{genSumm}, s_q \rangle$ for the query document d_q . A **base IMPerSumm decoder (b-Decoder)** feeds this directly into an internal pre-trained frozen model (decoder of a vanilla summarizer) for personalized summarization (task-3). However, $\mathbf{z}_b^{(t_{l+1})}$ lacks explicit contextualization of d_q , which can make it hard for the summarizer to decode (see Table 10). To address this, we extract the latent summary (s -node) embedding from $\mathbf{z}_b^{(t_{l+1})}$. **Latent s-Node Extraction** We reconstruct the s -node via a two-step inverse mapping. First, we recover the b -node embedding $\hat{\mathbf{e}}_{buj}^{(ti)} = W_k^+ \cdot \mathbf{z}_b^{(t_{l+1})}$, where W_k^+ is a learnable pseudo-inverse approximation kernel that disentangles the memory influence $\mathbf{m}_{uj}^{(t_{l+1})}$ from $\mathbf{z}_b^{(t_{l+1})}$ to approximate $\mathbf{e}_{buj}^{(t_{l+1})}$. We then infer the personalized latent summary embedding $\hat{\mathbf{e}}_{summary}^{(t_{l+1})} = W_{summary}^+ \cdot \hat{\mathbf{e}}_{buj}^{(ti)}$, $W_{summary}^+$ is a learnable

pseudo-inverse approximation kernel. While $\hat{\mathbf{e}}_{\text{summ}}^{(t_{l+1})}$ can also be fed to the internal frozen decoder (**s-Decoder**), it lacks the query document's representation completely.

Latent s-Node Contextualization. In contrast to the b-decoder and s-decoder, the IMPerSumm decoder contextualizes the extracted $\hat{\mathbf{e}}_{\text{summ}}^{(t_{l+1})}$ with the query document embedding using cross-attention. $\hat{\mathbf{e}}_{\text{summ}}^{(t_{l+1})}$ serves as the query, and the query document embedding $\mathbf{e}_d^{(t_l)}$ acts as both key and value, resulting in the contextualized latent summary embedding $\mathbf{e}_{\text{p-summ}}^{(t_{l+1})}$ as follows:

$$\mathbf{e}_{\text{p-summ}}^{(t_{l+1})} = \text{SoftMax} \left(\frac{(W_q \cdot \hat{\mathbf{e}}_{\text{summ}}^{(t_{l+1})})^\top (W_k \cdot \mathbf{e}_d^{(t_l)})}{\sqrt{d}} \right) \cdot W_v \cdot \mathbf{e}_d^{(t_l)} \quad (4)$$

where W_q , W_k and W_v are learnable projection matrices for the query, key, and value, respectively.

4.2.3 Task 3: Personalized Summarization. The document contextualized s-node $\mathbf{e}_{\text{p-summ}}^{(t_{l+1})}$ is fed into a pre-trained frozen decoder of a summarization model. We use the T5-large [34] decoder for this purpose.

Decoder Training. The decoder training objective ($\mathcal{L}_{\text{dec}}$) is a linear combination of two loss functions, *Generation Loss* ($\mathcal{L}_{\text{gen}}$) and the earlier encoder loss ($\mathcal{L}_{\text{enc}}$; see Section 4.2.1), as $\mathcal{L}_{\text{dec}} = \beta \cdot \mathcal{L}_{\text{gen}} + (1 - \beta) \cdot \mathcal{L}_{\text{enc}}$. Here $\mathcal{L}_{\text{gen}}$ is the cross-entropy loss between predicted tokens $\hat{y}$ and ground-truth y^* under teacher forcing with the frozen T5 decoder. Optimizing $\mathcal{L}_{\text{gen}}$ updates W_k , W_q , W_v and inverse-mapping weights W_{summ}^+ , W_k^+ . This ensures accurate latent s-node reconstruction $\hat{\mathbf{e}}_{\text{summ}}^{(t_{l+1})}$ and stronger cross-attention with document embedding $\hat{\mathbf{e}}_d^{(t_l)}$, thus improving summary relevance.

5 Evaluation

To evaluate IMPerSumm as an effective personalized summarizer, we investigate the three research questions discussed in Section 1.

5.1 Experiment Setup

Training Data. We construct UIGs from PENS, English-translated PersonalSum (via M2M-100 [10]), and OpenAI (Reddit) as $\mathcal{T}_{\text{train}}^{\text{P-D}}$, $\mathcal{T}_{\text{train}}^{\text{PS-EN}}$, and $\mathcal{T}_{\text{train}}^{\text{OAI}}$ to train IMPerSumm. We sample 52K trajectories from $\mathcal{T}_{\text{train}}^{\text{P-D}}$ (avg. 134 d -nodes, 5 s -nodes each), 21K from $\mathcal{T}_{\text{train}}^{\text{OAI}}$ (avg. 39 d -nodes, 10 s -nodes), and 700 from $\mathcal{T}_{\text{train}}^{\text{PS-EN}}$. Each trajectory is sliced before every (d - s) pair – forming training history $\tau_h^{u_j}$, query document d_q , and target summary s^{*u_j} .

Test Data. We prepare four test sets: $\mathcal{T}_{\text{test}}^{\text{P-D}}$, $\mathcal{T}_{\text{test}}^{\text{PS-EN}}$, and $\mathcal{T}_{\text{test}}^{\text{OAI}}$ for end-to-end personalized summarization, and $\mathcal{T}_{\text{test}}^{\text{M}}$ from MIND test data [46] to test sequential recommendation. For $\mathcal{T}_{\text{test}}^{\text{P-D}}$, each user's test trajectory τ^{u_j} , sampled from PENS stage-2 subset, is incrementally built by appending one (d - s) pair at a time to the stage-1 click history. At step k , $\tau_h^{u_j}$ includes all prior (d - s) pairs up to $k - 1$, and the k^{th} d -node serves as query d_q . We also insert 50–70 randomly sampled skipped d -nodes per user. For $\mathcal{T}_{\text{test}}^{\text{PS-EN}}$ and $\mathcal{T}_{\text{test}}^{\text{OAI}}$, we slice 500 and 3000 sampled trajectories, respectively, before each (d - s) pair. The test UIG $\mathcal{T}_{\text{test}}^{\text{M}}$ from MIND test data is created by augmenting the positive and negative targets as the tail-node of the last behavior with same history sequence (see Appendix C).

IMPerSumm Training. IMPerSumm is trained end-to-end over each user trajectory using $\mathcal{L}_{\text{dec}}$, with optimization carried out via the

Adam optimizer (PyTorch v2.0.1) at a learning rate of 1×10^{-3} . We train the full model for 10 epochs (approx. 30 hours) over 56K unique behavior sequences, using a batch size of 28. The decoder operates under teacher-forced supervision with a T5 backbone. $\mathcal{L}_{\text{dec}}$ enables the model to jointly learn dynamic behavior representations and generate summaries that are personalized, coherent, and faithful to both user history and source content.

5.2 Baseline Personalized Summarizers

(1) Personalized Summarizer Models. For RQ-1, we compare three SOTA personalized summarizers: PENS [2], GTP [36], and Signature-Phrase [3]. PENS couples a pointer-generator with external user encoders; we use the strongest off-the-shelf variants—NAML (T-1), EBNR, and NRMS (T-1/T-2) [29, 43, 44]. GTP integrates the TrRMIO user encoder; Signature-Phrase performs sequence modeling over user-specific keyphrases. Architecturally, NAML uses multi-view additive self-attention, EBNR is a GRU over click order, NRMS implements multi-head self-attention, and TrRMIO utilizes full-sequence transformer. All baselines are fine-tuned end-to-end for 2 epochs on $\mathcal{T}_{\text{train}}^{\text{P}}$.

(2) LLMs-as-summarizers. For RQ-2, we benchmark four **frozen** LLMs—Mistral-7B-Instruct [19], DeepSeek-R1-14B [7], LLaMA-2-13B-Chat-HF [37], and Zephyr-7B [38]—using the best 0-shot/2-shot prompts from Patel et al. [32] and prompt-chaining for DeepSeek-R1-14B and Mistral-7B-Instruct (details in App. E.2).

(3) Non-personalized Summarizers as Oracles. To further probe RQ-2, we evaluate BigBird-Pegasus [49] and SimCLS [27] following Vansh et al. [40]. We substitute each article's original title with the user's gold-reference title (cueing subjective preference). If exploited, these frozen models produce “personalized” summaries, effectively acting as oracles.

(4) News Recommendation Models. For RQ-3, we test IMPerSumm-encoder on next-interaction prediction and sequential recommendation. For next b -node prediction, we use PENS-based NAML/EBNR/NRMS; for recommendation, we compare against 23 sequential news recommenders, including DGAT [28], GNEWSREC [18], FastFormer [45], and MINER [1].

5.3 Personalization Evaluation Metrics

Automated Personalization Metric. To evaluate IMPerSumm as a personalized summarizer, we adopt PerSEval (PSE), the only metric explicitly designed for personalized summarization [6]. We report three PSE variants – PSE-SU4, PSE-METEOR, and PSE-JSD, each having strong human-judgment correlation and also computational efficiency. Details of PerSEval is in Appendix A.

Accuracy Evaluation. We also report the accuracy results of IMPerSumm based on Rouge-SU4 [23] and Rouge-L [24] metrics.

Encoder Evaluation. IMPerSumm's encoder is validated on next interaction prediction and sequential recommendation under RQ-3 using AUC, MRR, nDCG@5/10.

Human Judgment (HJ).

To validate IMPerSumm's human-alignment, we conducted a survey-based evaluation. In the survey, each participant² was shown a pair of gold-reference summary and a model-generated summary for

²Demography: 139 male, 32 female graduate students from Computer Science, Humanities, Mathematics, & Natural Sciences.

a specific news article from the PENS dataset. Model-generated summaries were drawn from IMPerSumm, along with six other best-performing baselines: PENS-NAML-T1, PENS-EBNR-T2, GTP, LLMs as DeepSeek-R1-14B and Mistral-7B under 2-shot w/history settings, and Bigbird-Pegasus as oracle. Model information was hidden to remove bias. Participants rated the model-generated summary similarity w.r.t the reference on a scale from 1 (low) to 6 (high). Each summary was rated by ≈ 4 respondents with a **low average inter-agreement** (Krippendorff- $\alpha = 0.3$), **thereby indicating subjectivity**. Hence, a higher average human rating with a low coefficient-of-variation (CV) would indicate better alignment of the model.

We also interpolate human ratings of our baseline models on the multi-domain non-news OpenAI-Reddit dataset. It contains multiple human-rated summaries of 9 models (≈ 6 raters/summary). We identify the top-rated (i.e., 7) one per user as the *human-preferred reference*. We then measure the RMSD-divergence of the model-generated summaries from the reference on the SBert-embedding-space. This leads to an average min-max range per rating map used for interpolation.

6 Results and Observations

In this section, we discuss the results in light of our three research questions. **All results have statistical significance $p < 0.01$.**

6.1 RQ-1: IMPerSumm Performance

IMPerSumm outperforms SOTA specialized personalized summarizers that ignore action distinctions, achieving **0.48/0.47/0.49 $\uparrow$** on PSE-JSD/SU4/METEOR over the strongest baseline GTP(+TrRMIO). This demonstrates the **effectiveness of explicit action-specific information modulation** via dynamic memory kernels over uniform RNN or transformer attention (see results in Table 1). When trained on the much smaller PersonalSum train UIG $\mathcal{T}_{\text{train}}^{\text{PS-EN}}$, it achieves **0.53/0.51/0.52** on PSE-JSD/SU4/METEOR on the PersonalSum test UIG $\mathcal{T}_{\text{test}}^{\text{PS-EN}}$, underscoring its effectiveness on tailored datasets. IMPerSumm also attains average boost **48.9/43.6 $\uparrow$** in terms of RG-SU4 and RG-L accuracy metrics. IMPerSumm also excels in survey-based HJ evaluation (CV = 0.08) despite inherent subjectivity, indicating stronger personalization, as well as in interpolated human ratings with low avg. RMSD from the references (Table 2). **Ablation: Effect of Memory Kernels.** To assess action-specific encoding, we ablate IMPerSumm’s encoder with a fixed cross-attention decoder in three settings: (i) base model w/o MI, (ii) with KDE-MI (MI Encoder), and (iii) KDE-MI w/ adaptively fused memory kernels (MI+S as MI w/ m_{short} , MI+SL as MI w/ m_{short} and m_{long} , MI+SLE as MI w/ m_{short} , m_{long} , and m_{event} or IMPerSumm-Full). The base model using cosine similarity performs worst, showing simplistic embedding-level similarity fails to capture preference shifts. Combining MI with m_{short} and m_{long} (MI+SL) achieves **97/96/95%** of IMPerSumm-Full performance, confirming that action-aware modulation and dynamic memory kernels are critical.

Ablation: Effect of Contextualized Injection. In order to understand the effect of contextualized injection, we use the best-performing MI+SLE encoder and compare *b*-node injection (**b-Decoder**), reconstructed *s*-node injection w/o document-context (**s-Decoder**), and w/ document-context (**IMPerSumm-Full**). We find

Table 1: RQ-1: Personalized Summarization on PENS (20k test docs). Baselines show reported means; IMPerSumm variants show *mean $\pm$ 95% CI* from three random resampling trials (25%, 50%, 75% subsets). IMPerSumm (Full) achieves 47–55% relative gain over the best baseline.

Category	Model / Variant	PSE-JSD	PSE-SU4	PSE-METEOR
Personalized Summarizers	PENS-NAML-T1	0.021	0.014	0.016
	PENS-EBNR-T1	0.015	0.010	0.011
	PENS-EBNR-T2	0.011	0.008	0.009
	PENS-NRMS-T1	0.015	0.011	0.011
	PENS-NRMS-T2	0.008	0.007	0.007
	GTP	0.024	0.017	0.019
	SP-Individual	0.017	0.015	0.014
IMPerSumm (Encoder Ablations)	Base Encoder + CA-Decoder	0.241 $\pm$ 0.02	0.013 $\pm$ 0.01	0.107 $\pm$ 0.02
	MI-Encoder + CA-Decoder	0.311 $\pm$ 0.02	0.054 $\pm$ 0.01	0.274 $\pm$ 0.02
	MI+S-Encoder + CA-Decoder	0.357 $\pm$ 0.02	0.056 $\pm$ 0.01	0.334 $\pm$ 0.02
	MI+L-Encoder + CA-Decoder	0.387 $\pm$ 0.02	0.265 $\pm$ 0.02	0.402 $\pm$ 0.02
	MI+SL-Encoder + CA-Decoder	0.502 $\pm$ 0.02	0.483 $\pm$ 0.03	0.491 $\pm$ 0.03
	MI+LE-Encoder + CA-Decoder	0.485 $\pm$ 0.02	0.436 $\pm$ 0.02	0.463 $\pm$ 0.03
	MI+SE-Encoder + CA-Decoder	0.462 $\pm$ 0.02	0.447 $\pm$ 0.02	0.458 $\pm$ 0.02
IMPerSumm (Decoder Ablations)	MI+SLE-Encoder + s-Decoder	0.325 $\pm$ 0.02	0.096 $\pm$ 0.01	0.202 $\pm$ 0.01
	MI+SLE-Encoder + b-Decoder	0.331 $\pm$ 0.02	0.106 $\pm$ 0.01	0.237 $\pm$ 0.02
	MI+SLE-Encoder + CA-Decoder (Full)	0.518$\pm$0.02	0.501$\pm$0.03	0.514$\pm$0.03

Table 2: Qualitative Comparison w.r.t ROUGE, Survey-based Human Eval (PENS Dataset; Krippendorff- $\alpha = 0.3$; CV: Co-efficient-of-Variation), & Human Ratings (OpenAI-Reddit).

Model Name	RG-SU4	RG-L	Direct Human Eval	RMSD	Interpolated Human Rating
IMPerSumm-Full	63.61	66.51	3.15 (CV=0.08)	0.34	7
Mistral-7B	16.42	22.85	<u>3.05</u> (CV = 0.07)	0.79	5
BigbirdPegasus	17.13	21.65	<u>3.02</u> (CV = 0.08)	0.81	5
DeepSeek-14B	19.57	<u>29.72</u>	<u>3.0</u> (CV = 0.08)	0.78	5
PENS-EBNR-T2	12.41	20.82	2.97 (CV = 0.11)	0.93	2
PENS-NAML-T1	13.12	21.62	2.94 (CV = 0.09)	0.92	2
GTP	<u>21.91</u>	28.31	<u>2.86</u> (CV = 0.09)	0.94	2

that IMPerSumm-full outperforms b-Decoder by an avg. of **0.27 $\uparrow$** and s-Decoder by an avg. of **0.29 $\uparrow$** w.r.t. PSE variants; see Table 1.

6.2 RQ-2: Effectiveness of Preference Injection as Prompts & Cues

We study whether personalization can be achieved without a dedicated behavioral encoder, either through history-aware LLM prompting or as cue-injection into vanilla baseline summarizers (thereby testing their limit as oracles).

RQ-2(a): History-aware LLM Prompting. While injecting user histories as prompts in LLMs shows improvements in personalization over traditional models [32], we assess its ability to replace a dedicated behavioral encoder by prompting LLMs with user histories (Section 5.2) via 0-shot, 2-shot+history, and prompt-chaining. IMPerSumm outperforms these approaches by **0.33/0.42/0.40 $\uparrow$** on 0/2-shot and **0.44/0.47/0.49 $\uparrow$** across PSE on prompt chaining across four LLMs, indicating that mid-sized LLM prompting techniques are not as effective.

Ablation: Cross-domain Generalizability. We train IMPerSumm on OpenAI-Reddit $\mathcal{T}_{\text{train}}^{\text{OAI}}$ and evaluate on $\mathcal{T}_{\text{test}}^{\text{OAI}}$ (Section 5.1) for cross-domain generalizability³. IMPerSumm outperforms the best LLM baseline, DeepSeek-R1 by **0.33/0.36/0.38 $\uparrow$** , across PSE variants showing cross-domain robustness. Results are in Table 4.

RQ-2(b): Oracle Summarizers with Gold References. We evaluate oracle versions of BigBird-Pegasus and SimCLS to analyze whether they can serve as strong baselines. Despite having access to gold cues, IMPerSumm outperforms BigBird-Pegasus by **0.29 $\uparrow$** and

³OAI covers 29 wide-ranging non-news topics with greater diversity (App. C).

Table 3: Performance comparison on the PENS dataset across LLM variants and IMPerSumm under RQ-2.

Category	Model	PSE-JSD	PSE-SU4	PSE-METEOR
LLMs (0-shot history)	LLaMA-13B	0.187	0.069	0.078
	Mistral-7B	0.212	0.082	0.098
	DeepSeek-14B	0.152	0.078	0.084
	Zephyr-7B	0.211	0.081	0.089
LLMs (2-shot history)	LLaMA-13B	0.227	0.078	0.081
	Mistral-7B	0.235	0.087	0.084
	DeepSeek-14B	0.248	0.094	0.097
	Zephyr-7B	0.231	0.085	0.086
LLMs (Prompt-chaining)	Mistral-7B	0.072	0.026	0.023
	DeepSeek-14B	0.078	0.028	0.024
Vanilla (Cue Injection)	BigbirdPegasus	0.253	0.143	0.168
	SimCLS	0.157	0.032	0.116
IMPerSumm-Full	MI+SLE-Encoder+CA-Decoder	0.518	0.501	0.514

Table 4: RQ2: Performance comparison on the OpenAI (Red-dit) dataset across LLMs w/ 2-shot user history and IMPerSumm.

Category	Model	PSE-JSD	PSE-SU4	PSE-METEOR
LLMs (2-shot history)	LLaMA-13B	0.232	0.093	0.107
	Mistral-7B	0.226	0.088	0.103
	DeepSeek-14B	0.243	0.095	0.109
	Zephyr-7B	0.214	0.087	0.104
IMPerSumm-Full	MI+SLE-Encoder+CA-Decoder	0.576	0.454	0.494

Table 5: RQ-3 (a): Next b-node Prediction (PENS Dataset). † NT: Originally published results were on the news recommendation task in contrast to next behavior prediction. For IMPerSumm, values show *mean* $\pm$ 95% CI; Setup as in Table 1.

Category	Model	AUC	MRR	nDCG@5	nDCG@10
User Encoders (End-to-end Trained on Summarization)	NAML†	0.498	0.001	0.0004	0.0007
	NRMS†	<u>0.499</u>	0.0009	0.0002	0.0004
	EBNR†	<u>0.499</u>	0.0009	0.0003	0.0005
IMPerSumm	MI+SL-Encoder	0.431 $\pm$ 0.035	<u>0.106</u> $\pm$ 0.015	<u>0.073</u> $\pm$ 0.015	<u>0.117</u> $\pm$ 0.020
	MI+SLE-Encoder	0.513 $\pm$ 0.042	0.216 $\pm$ 0.030	0.181 $\pm$ 0.005	0.236 $\pm$ 0.017

SimCLS by **0.44**↑ across PSE variants JSD/SU4/METEOR. This underscores the effectiveness of IMPerSumm’s memory-gated, action-specific encoding (detailed results in Table 3).

6.3 RQ-3: IMPerSumm-Encoder Validation

We validate IMPerSumm’s encoder along two axes—*Next b-node Prediction* and *Sequential Recommendation* under the hypothesis that better next-interaction prediction and sequential ranking by user-encoder are expected so as to align with the observed higher personalized summarization performance.

RQ-3(a): Next b-Node Prediction. We see that user-encoder baselines (NAML/NRMS/EBNR), end-to-end trained on the PENS dataset on the summarization task along with the PENS-framework decoder, perform near-random on this task (MRR/nDCG $\approx$ 0). Using an *action-specific* MI encoder with multi-horizon memory, MI+SL (short+long kernels) improves performance (0.431 AUC, 0.106 MRR, 0.073/0.117 nDCG@5/10), while MI+SLE (adds event kernel) achieves the best results (0.513 AUC, 0.216 MRR, 0.181/0.236 nDCG@5/10), showing that action-conditioned MI combined with short, long, and event memory better captures evolving user (see Table 5).

RQ-3(b): Sequential Recommendation. We use the standard and widely adopted MIND news recommendation test benchmark.

Table 6: RQ-3(b): Sequential Recommendation on MIND-Large. (sorted by MRR) Baselines show paper-reported means. IMPerSumm results denote *mean* $\pm$ 95% CI; Setup as in Table 1.

Methods (Venue, Year)	AUC	MRR	nDCG@5	nDCG@10
DKN (WWW’18)	64.07	30.42	32.92	38.66
EBNR (KDD’17)	65.46	31.26	32.18	39.04
NPA (KDD’19)	65.92	32.07	34.72	40.37
NAML (IJCAI’19)	66.46	32.75	35.66	41.40
LSTUR (ACL’19)	67.08	32.86	35.95	40.94
Poolingformer (ICML’21)	68.54	33.20	36.69	42.60
NRMS (EMNLP-IJCNLP’19)	67.66	33.25	36.28	41.98
BigBird (NeurIPS’20)	68.14	33.28	36.42	42.18
Transformer (NeurIPS’17)	68.22	33.32	36.35	42.23
GERL (WWW’20)	68.10	33.41	36.34	42.03
GNewsRec (IP&M’20)	68.15	33.45	36.43	42.10
FIM (ACL’20)	67.87	33.46	36.53	42.21
HieRec (ACL-IJCNLP’21)	68.33	33.86	36.83	42.65
DCAN (arXiv’22)	68.90	33.90	36.90	42.80
ANRS (arXiv’22)	69.20	34.10	37.10	43.00
TCCM (CIKM’23)	69.75	34.42	37.53	43.25
Fastformer (arXiv’21)	69.11	34.55	37.62	43.38
FUM (SIGIR’22)	69.90	34.60	37.70	43.40
CAUM (SIGIR’22)	70.04	34.71	37.89	43.57
DIGAT (Findings ACL’22)	70.08	35.20	38.46	44.15
PLM-NR (SIGIR’21)	70.64	35.39	38.71	44.38
Fastformer+PLM-NR (Hybrid)	71.04	35.91	39.16	45.03
MINER (Findings ACL’22)	71.51	36.06	39.56	45.21
CAST-Rec (TOIS’25)	<u>72.10</u>	36.90	40.20	46.30
Fastformer+PLM-NR-Ensemble (Hybrid’22)	72.68	<u>37.45</u>	<u>41.51</u>	46.84
MI+SL-Encoder (IMPerSumm)	48.13 $\pm$ 0.7	<u>37.48</u> $\pm$ 0.4	40.81 $\pm$ 0.5	<u>50.86</u> $\pm$ 0.6
MI+SLE-Encoder (IMPerSumm)	55.62 $\pm$ 0.6	40.03 $\pm$ 0.6	45.07 $\pm$ 0.7	53.42 $\pm$ 0.7

Since the PENS train set is derived from the MIND train set, this test is **cross-task** rather than cross-domain. We observe that the PENS-trained $\mathcal{T}_{\text{test}}^M$, MI+SLE-Encoder improves head-of-list ranking over the best baseline (Fastformer+PLM-NR-Ensemble): MRR 2.58 ↑, nDCG@5 3.56 ↑, nDCG@10 6.58 ↑ (see Table 6). Lower AUC ($\approx$ 16 ↓) indicates that although the IMPerSumm-encoder develops cross-task capability of news recommendation when exclusively trained on personalized summarization, it requires explicit training on the recommendation task for a corresponding *classification* ability.

7 Conclusion

In this paper, we propose IMPerSumm, a personalized text summarizer that models user behavior as dynamic information modulation per interaction. It encodes sequences via feature-wise mutual information and aggregates memories with learnable temporal kernels to capture evolving preferences. Unlike static profiles or prompt-based methods, it offers interpretable adaptation with gains in relevance, semantic alignment, and preference sensitivity. We see 55% gain in PSE w.r.t SOTA baselines. As future work, we are exploring adaptive kernels, advanced encoding, decoder-side personalization, and alignment reward-based reinforcement objectives.

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A Measuring Degree of Personalization

Dasgupta et al. [6] proposed PerSEval as arguably the only metric for direct evaluation of degree-of-personalization. It is an extension to DEGRESS that measures the proportionate degree of divergence (**responsiveness**) of model-generated summaries w.r.t the subjective difference in the expected summaries of the users [40]. PerSEval penalizes the lack of personalization while also accounting for accuracy. Given a document d_i and user j , summary-level PerSEval is:

$$\text{PerSEval}(s_{u_{ij}}|(d_i, u_{ij})) = \text{DEGRESS}(s_{u_{ij}}|(d_i, u_{ij})) \times \text{EDP}(s_{u_{ij}}|(d_i, u_{ij})) \quad (5)$$

where DEGRESS measures responsiveness and EDP is the Effective DEGRESS penalty that incorporates both accuracy-drop and accuracy-inconsistency on degree-of-responsiveness. The system-level score averages over all (i, j) .

PSE-SU4 uses ROUGE-SU4 skip-bigram F1 as σ , and is shown to correlate strongly with human judgment. On the other hand, **PSE-JSD** uses Jensen–Shannon Divergence between n -gram distributions of generated vs. reference summaries, while **PSE-Meteor** uses $1 - \text{METEOR}(G, R)$, where METEOR combines unigram matches with synonym/paraphrase handling.

B Information Modulation via KDE

We estimate how much information a current user action (e.g., a click or a generated summary) retains from its immediate past by computing the **mutual information (MI)** between their embeddings via **Kernel Density Estimation (KDE)**. Each behavior step $b_{u_j}^{(t_i)}$ consists of a pair of d -dimensional embeddings $(\mathbf{c}_{hd}^{(t_{i-1})}, \mathbf{e}_{tl}^{(t_i)})$, where $\mathbf{c}_{hd}^{(t_{i-1})}$ represents the head-cell content (historical context) and $\mathbf{e}_{tl}^{(t_i)}$ the tail-node embedding (current action). High MI indicates *continuity*—the current action is predictable from the user’s prior behavior—while low MI suggests *novelty* or *drift*. The modulation process thus quantifies how information flows from the past to the present along a user’s trajectory τ^{u_j} .

KDE-based density estimation. Following the KDE formulation, the marginal densities of the head and tail embeddings

are given by: $\hat{p}_{hd}(\mathbf{c}_{hd}) = \frac{1}{(i-1)\lambda^d} \sum_{j=1}^{i-1} K\left(\frac{\mathbf{c}_{hd} - \mathbf{c}_{hd}^{(t_j)}}{\lambda}\right)$ & $\hat{p}_{tl}(\mathbf{e}_{tl}) = \frac{1}{(i-1)\lambda^d} \sum_{j=1}^{i-1} K\left(\frac{\mathbf{e}_{tl} - \mathbf{e}_{tl}^{(t_j)}}{\lambda}\right)$. and the joint density between the current tail-node and accumulated history is:

$$\hat{p}_{hd,tl}(\mathbf{c}_{hd}, \mathbf{e}_{tl}) = \frac{1}{(i-1)\lambda^{2d}} \sum_{j=1}^{i-1} K\left(\frac{\mathbf{c}_{hd} - \mathbf{c}_{hd}^{(t_j)}}{\lambda}\right) K\left(\frac{\mathbf{e}_{tl} - \mathbf{e}_{tl}^{(t_j)}}{\lambda}\right). \quad (6)$$

Table 7: Symbol and Notation Table for IMPerSumm

Symbol	Description
u_j	The j -th user
$d^{(t_p)}$	Document (d-node) interacted with at time-step t_p
$s_j^{(t_q)}$	Summary (s-node) generated/read by user u_j at time t_q
$a^{(t_p)}$	User action at time-step t (e.g., click, skip, summarize)
τ^{u_j}	Trajectory of user u_j : sequence of interactions over time
$\tau_b^{u_j}$	Behavior trajectory of user u_j over the b-tier graph
$G = (N, E)$	User-Interaction Graph (UIG) with node set N and edge set E
$b_{u_j}^{(t_i)}$	Behavior triple at time t_i : (head, $a^{(t)}$, tail)
$e_{hd}^{(t_i)}$	Embedding of the head node at time-step t_i
$e_{tl}^{(t_i)}$	Embedding of the tail node at time-step t_i
$I^{(t_i)}$	Mutual information at time-step t_i
$c_{hd}^{(t_i)}$	Content Embedding of the head node at time-step t_i
$c_{tl}^{(t_i)}$	Content Embedding of the tail node at time-step t_i
f_{cos}	Cosine-based alignment function
$g^{(clk, t_i)}$	Learnable Click gate
$g^{(skp, t_i)}$	Learnable forget gate for skip interactions
$g^{(f, t_i)}$	Learnable focus gate for summarize (genSumm) actions
$g^{(summ, t_i)}$	Gating parameter for incorporating influential history in summGen
$g^{(acc, t_i)}$	Gating parameter for document relevance in summGen
$m_{short}^{(t)}$	Short-term memory at time-step t
$m_{long}^{(t)}$	Long-term memory at time-step t
$m_{event}^{(t)}$	Event-specific memory at time-step t
$\mathcal{K}_{short}, \mathcal{K}_{long}, \mathcal{K}_{event}$	Learnable kernels for temporal memory aggregation
$g_{fuse}^{(t_i)}$	Time-variant gating vector for fusing memories
$z_b^{(t_i)}$	Memory-infused Behavior embedding t
$\tilde{e}_{summ}^{(t_i)}$	Latent summary vector w/o cross-attention
$e_{p-summ}^{(t_i)}$	Document contextualized Latent summary vector from cross-attention
d_q	Query document
s_q^*	Ground-truth personalized summary for document d_q

Here $K(\cdot)$ is a Gaussian kernel defined as: $K(\mathbf{u}) = \exp(-\frac{1}{2}\|\mathbf{u}\|^2)$, which assigns higher weights to semantically proximate embeddings. The bandwidth λ controls the smoothness of the estimation: a smaller λ emphasizes fine-grained local continuity, whereas a larger λ smooths over broader historical variation. In embedding space, K thus captures semantic affinity, and λ determines the scale of historical context considered.

Kernelized MI estimate. Using these densities, we estimate the mutual information between the head-cell and the tail-node embeddings as:

$$I(c_{hd}; e_{tl}) = \frac{1}{(i-1)} \sum_{j=1}^{i-1} \log \frac{\hat{p}_{hd,tl}(c_{hd}^{(t_j)}, e_{tl}^{(t_j)})}{\hat{p}_{hd}(c_{hd}^{(t_j)}) \hat{p}_{tl}(e_{tl}^{(t_j)})}. \quad (7)$$

This MI score captures the degree of *information modulation*—how much of the present action’s embedding is predictable from prior user context. We fix $\lambda = 0.6$, obtained via Silverman’s rule-of-thumb [35] and validated on the development set, to maintain stable yet locality-aware estimation.

C Datasets

PENS Dataset The PENS dataset [2] contains 113,762 news articles across 15 topics, with titles (avg. 10.5 words), bodies (avg. 549 words), and WikiData-linked entities. It also includes user interaction logs derived from MIND [46]. The **training set** comprises 500k impressions sampled (June 13–July 3, 2019), recording user IDs, clicked/unclicked news, and click histories. These logs highlight strong preference shifts, underscoring temporal personalization. The **test set** has 103 students who reviewed 1,000 headlines, selected 50, and created personalized headlines for 200 unseen articles.

Four annotators per article ensured coverage, and editors filtered inaccuracies. The remaining gold headlines serve as personalized references.

OpenAI (Reddit) Dataset The OpenAI Reddit dataset [41] includes 123,169 posts from 29 subreddits. It offers OpenAI- and human-written summaries, split into Comparisons (training/validation) and Axis (validation/testing). A curated set of 1,038 posts produced 7,713 summaries, later rated by 64 annotators on accuracy, coherence, coverage, and quality.

PersonalSum Dataset PersonalSum [50] provides 441 Norwegian news articles and 1,099 personalized summaries from 39 annotators. Each article has a GPT-4-generated generic summary, plus 3 personalized summaries reflecting annotator preferences. Annotators highlight source text, and quality checks (GPT-3.5 scoring) ensure quality user-grounded outputs.

UIG Construction

We construct User-Interaction Graphs (UIGs) from two dataset types: (i) trajectory-based (e.g., PENS [2]), where clicks/skips form interaction sequences, and (ii) feedback-based (e.g., OpenAI-Reddit [41]), where posts and summaries are linked by user ratings. For PENS, we enrich click/skip logs with summary nodes from the test set, yielding $\mathcal{T}^{P-D}$. For Reddit, we treat posts as clicked if any summary scored $\geq 6/9$, selecting the top-rated summary as its surrogate node.

MIND dataset UIG For sequential news recommendation experiments, we construct the test UIG $\mathcal{T}_{test}^M$ using the standard MIND-large test data [46]. Given a user trajectory in MIND, we first extract the last observed behavior sequence, which serves as the user’s test history. To this history, we augment both the positive and negative targets provided in the MIND test set where **positive targets** (appended as *tail nodes* to the user’s history sequence) are the news articles that the user actually clicked during the test phase, while **negative targets** (appended as *tail nodes*) are the candidate news articles that the user was exposed to but did not click (sampled negatives in MIND). This augmentation produces a unified test UIG where each user’s final behavior history points to multiple candidate tail nodes—some positive (clicks) and others negative (non-clicks). Such a structure is essential for evaluating ranking-based sequential recommendation, since the model must differentiate relevant (positive) from irrelevant (negative) targets given the same history context.

D Baselines

Baseline LLMs:

- 1. Zephyr 7B- β** [38] is a 7B transformer fine-tuned from Mistral-7B using DPO. It relaxes alignment constraints to boost raw performance, achieving strong MT-Bench scores, and is openly released under MIT license.
- 2. Mistral-Instruct 7B** [19] is a dense transformer with GQA and SWA, trained on 2T tokens. It outperforms larger models like LLaMA2-13B in many benchmarks.
- 3. LLaMA-2-13B** [37] is a 13B transformer trained on 2T tokens with RLHF-based chat tuning. It remains widely used though surpassed by smaller models such as Mistral 7B.

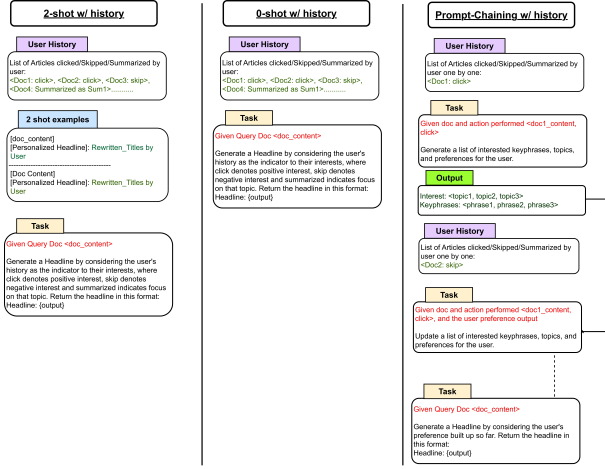


Figure 5: Prompt-Templates for LLM baselines.

4. **DeepSeek-R1 14B** [7] is a 14.8B distilled model optimized for math, code, and reasoning. Despite its size, it matches larger models on AIME/MATH benchmarks and is MIT licensed.

Baseline Generic (Oracle) Summarizers:

1. **BigBirdPegasus** [49] extends Pegasus with sparse, global, and random attention, enabling efficient long-sequence summarization.
2. **SimCLS** [27] trains a Seq2Seq generator with MLE and a RoBERTa-based contrastive ranker to refine candidate summaries.

Baseline Personalized Models

1. **PENS Framework** uses: (i) NRMS [44] that encodes titles and user clicks with self-attention. In Type 1, user embeddings initialize the decoder; in Type 2, they modulate attention weights; (ii) NAML [43] that encodes multiple news views (titles, bodies, topics). Its user embedding personalizes decoding via Injection-Type 1; (iii) EBNR [29] that uses GRU-based browsing histories for embeddings. Type 1 initializes the decoder, while Type 2 personalizes via attention layers.

2. **General Then Personal (GTP)** GTP [36] first trains a general headline generator, then applies a user-customizer with control codes. ISB and MUM modules ensure fidelity and robust personalization.

3. **Signature Phrase** Signature Phrases [3] distill histories into dynamic phrase-level profiles via contrastive learning. These high-level signals guide personalized headline generation.

E Implementation Details

E.1 Compute Resources

All preprocessing (graph construction, embedding lookup, probability mapping) was done on CPUs with 16GB/core. Embedding tables for bodies, headlines, and summaries were seeded from pretrained E5 embeddings [42] and applied 192-D PCA over all embeddings. Training and inference for IMPerSumm used mixed-precision on A10 and L40S GPUs, provided by Lightning.ai.

Table 8: Complete list of hyperparameters and their shapes. Includes classification weights and s/d cross-attention.

Name	Symbol	Dimension
Head-cell weight matrix	W_h	$\mathbb{R}^{192 \times 192}$
Head-node embedding weight	W_{hd}	$\mathbb{R}^{192 \times 192}$
Tail-cell weight matrix	W_a	$\mathbb{R}^{192 \times 192}$
b-node output weight	W_b	$\mathbb{R}^{192 \times 192}$
b-node output bias	b_b	$\mathbb{R}^{192}$
Cell content (head/tail)	$c_{hd}^{(t_i)}, c_{td}^{(t_i)}$	$\mathbb{R}^{192}$
Cell content (head/tail)	$e_{bu_j}^{(t_i)}$	$\mathbb{R}^{192}$
Cosine modulation influence factor	λ_c	Scalar
Cosine modulation influence factor	$c_{hd}^{(a, t_i)}$	$\mathbb{R}^{192}$
Mutual information function	$I_K(\cdot; \cdot)$	$\mathbb{R}^{192}$
Click gate	$g^{(clk, t_i)}$	$\mathbb{R}^{192}$
Click gate weight	W_g^{clk}	$\mathbb{R}^{192 \times 192}$
Skip gate	$g^{(skp, t_i)}$	$\mathbb{R}^{192}$
Skip gate weight	W_g^{skp}	$\mathbb{R}^{192 \times 192}$
Focus gate	$g^{(fc, t_i)}$	$\mathbb{R}^{192}$
Focus gate weight	W_g^{fc}	$\mathbb{R}^{192 \times 192}$
Interest gate (summary)	$g^{(summ, t_i)}$	$\mathbb{R}^{192}$
Interest gate weight	$W_{summ}^{(g)}$	$\mathbb{R}^{192 \times 192}$
Accuracy gate (summary)	$g^{(acc, t_i)}$	$\mathbb{R}^{192}$
Accuracy gate weight	$W_{acc}^{(g)}$	$\mathbb{R}^{192 \times 192}$
Short-term memory kernel weight	W_{short}	$\mathbb{R}^{192}$
Long-term memory kernel weight	W_{long}	$\mathbb{R}^{192}$
Event-specific memory projection	W_{event}	$\mathbb{R}^{192 \times 192}$
Short-term memory vector	$m_{short}^{(t_i)}$	$\mathbb{R}^{192}$
Long-term memory vector	$m_{long}^{(t_i)}$	$\mathbb{R}^{192}$
Event-specific memory vector	$m_{event}^{(t_i)}$	$\mathbb{R}^{192}$
Short-term kernel value	$\mathcal{K}_{short}^{t_j}$	$\mathbb{R}^{192}$
Long-term kernel value	$\mathcal{K}_{long}^{t_j}$	$\mathbb{R}^{192}$
Event-specific kernel value	$\mathcal{K}_{event}^{t_j}$	$\mathbb{R}^{192 \times 3}$
Memory fusion gate weight	W_{fuse}	$\mathbb{R}^3$
Memory fusion gate value	$g_{fuse}^{(t_i)}$	$\mathbb{R}^3$
Fused memory vector (before gating)	$m_{fuse}^{(t_i)}$	$\mathbb{R}^3$
Gated memory output	$m_{u_j}^{(t_i)}$	$\mathbb{R}^3$
Residual vector	$c_{td}^{(t_i)}$	$\mathbb{R}^{192}$
Final memory-infused b-node embedding	$z_b^{(t_i)}$	$\mathbb{R}^{192}$
Cross-attention query projection	W_Q	$\mathbb{R}^{192 \times 192}$
Cross-attention key projection	W_K	$\mathbb{R}^{192 \times 192}$
Cross-attention value projection	W_V	$\mathbb{R}^{192 \times 192}$
Cross-attention output projection	W_O	$\mathbb{R}^{192 \times 192}$
History b-node classifier	$W_{cls-hist}$	$\mathbb{R}^{76,000 \times 192}$
Next b-node classifier	$W_{cls-next}$	$\mathbb{R}^{76,000 \times 192}$
s-Node Approximator	W_{summ}^+	$\mathbb{R}^{192 \times 192}$

E.2 Prompt Details

2-shot w/ history. This setup presents the model with the user's full interaction history of clicks, skips, and summaries. Two in-context examples, each pairing an article with the user's rewritten headline, are provided before the task. Using these demonstrations together with the history, the model generates a personalized headline for the new document.

0-shot w/ history. Here, the model again receives the complete interaction history containing clicks, skips, and summaries. However, no examples are shown; instead, the prompt explains the meaning of each action type. The model must rely on zero-shot reasoning over this history to produce the personalized headline.

Prompt-Chaining w/ history. This method structures personalization as a step-by-step process. For each document and action, the model extracts topics, keyphrases, and inferred preferences, updating a running profile. When the query document appears, the accumulated profile is used to generate the personalized headline.